Requirements Specification

Delivery Reliability Monitoring for an Online Marketplace

Companion to the Business Requirements Document (BRD version 1.2)

Item	Detail
Document status	Final, version 1.0
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Related document	Business Requirements Document, Delivery Reliability Monitoring, version 1.2
Note	Practice case study using public data (Olist, Kaggle, CC BY-NC-SA 4.0). Stakeholders, targets and thresholds are assumed.

1. Purpose

The BRD says what the business needs. This document breaks each business requirement (BR) down into three things: functional requirements (FR), which say what the dashboard must do; non-functional requirements (NFR), which say how well it must work; and user stories with acceptance criteria, which say how each need is tested.

The solution is an interactive dashboard, supported by an Excel and SQL analysis, that lets the operations team track delivery performance.

2. Conventions and Definitions

Wording. 'Shall' marks a requirement that must be met. Priority uses MoSCoW (Must, Should, Could). Verification: T = tested against the acceptance criteria, I = checked by looking at it, R = numbers compared with the Excel and SQL analysis.

ID	Definition
D-01	Delivery base: orders with status delivered and a delivery date to the customer (96,470 orders)
D-02	Late order: delivered after the estimated delivery date shown at purchase (BRD rule BRU-01)
D-03	Delay buckets: on time, 1-3, 3-7, 7-14 and over 14 days late (BRD rule BRU-03)
D-04	Reviewed order: an order with a review. Where there are several reviews, the first is used (BRU-04)
D-05	Item revenue: sum of item price, excluding freight. Canceled and unavailable orders are excluded (BRU-06)
D-06	Same-state order: the customer and the seller of the order's first item (order_item_id = 1) are in the same state. Orders with items from several sellers are grouped by that first item only, which is a simplification
D-07	Overall baseline: dashboard values with no filter applied

3. Functional Requirements

ID	The dashboard shall...	Priority	Source	Verification
FR-01	Show the late rate and on-time rate for the delivery base as headline figures	Must	BR-01	T, R
FR-02	Show average, median and 90th percentile delivery days as headline figures	Must	BR-02	T, R
FR-03	Show the number of orders in the delivery base next to the headline figures	Must	BR-01, BR-10	T, R

ID	The dashboard shall...	Priority	Source	Verification
FR-04	Show late rate and order count by purchase month for January 2017 to August 2018	Must	BR-03	T, R
FR-05	Show the overall late rate as a reference on the monthly trend so peaks stand out	Should	BR-03	I
FR-06	Show late rate and average delivery days by customer state	Must	BR-04	T, R
FR-07	Compare late rate and delivery days for same-state and cross-state orders (D-06)	Must	BR-04	T, R
FR-08	List sellers with at least 30 delivered orders, showing orders, late rate and average review score, sortable by late rate	Should	BR-05	T, R
FR-09	Show average review score and order count for each delay bucket, and for late versus on-time orders	Must	BR-06	T, R
FR-10	Show late rate, orders and item revenue by product category using English names, with missing categories shown as Unknown	Should	BR-07	T, R
FR-11	Show order count and item revenue by month, with freight shown separately	Should	BR-08	T, R
FR-12	Provide filters for purchase month range, customer state and product category that apply to every view	Must	BR-09	T
FR-13	Provide a definitions and limitations page listing metric definitions, known data issues and the number of orders excluded	Must	BR-10	I

4. Non-Functional Requirements

The limits below are my assumptions, because this is a practice case study.

ID	Requirement	Priority	Verification
NFR-01	Accuracy: counts in the dashboard shall equal the Excel and SQL analysis exactly, and averages shall match to two decimal places	Must	R
NFR-02	Consistency: Excel, SQL and the dashboard shall use the definitions in section 2	Must	R
NFR-03	Performance: each page and each filter change shall respond in under 3 seconds on a standard laptop	Should	T
NFR-04	Usability: a first-time user shall find the overall late rate within 10 seconds	Should	T
NFR-05	Readability: colour shall not be the only way meaning is shown, and every chart shall have a title and labelled axes	Should	I
NFR-06	Maintainability: refreshing the data shall require only replacing the source files and refreshing, with the steps documented	Could	I
NFR-07	Privacy: the solution shall use only the anonymised identifiers in the source data and shall not try to identify individuals	Must	I
NFR-08	Traceability: every measure shall have a written definition and every requirement shall map to a view or to a recorded gap (section 8)	Must	I

5. User Stories and Acceptance Criteria

Format: As a [role], I want [capability], so that [benefit]. Acceptance criteria use Given, When, Then. Figures are the overall baseline (D-07) calculated from the source data.

US-01 Headline performance (FR-01, FR-02, FR-03)

As the Head of Operations, I want to see the late rate and delivery time headline figures, so that I know how delivery is performing at a glance.

- Given no filters are applied, when I open the overview, then the delivery base shows 96,470 orders.
- Given no filters are applied, then the late rate shows 8.1% and the on-time rate shows 91.9%.
- Given no filters are applied, then average delivery time shows 12.56 days, the median 10.22 days and the 90th percentile 23.1 days.

US-02 Monthly trend (FR-04, FR-05)

As the Head of Operations, I want to see the late rate by month, so that I can spot peak periods.

- Given no filters are applied, when I view the trend, then 20 months from January 2017 to August 2018 are shown.
- Then the three months with the highest late rate are March 2018 (21.4%), February 2018 (16.0%) and November 2017 (14.3%).
- Then the overall late rate is visible as a reference line.

US-03 Regional performance (FR-06)

As the Logistics Partner Manager, I want delivery performance by customer state, so that I can see where late deliveries concentrate.

- Given no filters are applied, when I view the state breakdown, then all 27 customer states are shown.
- Then the orders across all states add up to 96,470.
- When I select a state, then the other views update to that state only.

US-04 Same-state versus cross-state (FR-07)

As the Logistics Partner Manager, I want to compare same-state and cross-state orders, so that I know whether distance drives lateness.

- Given no filters are applied, when I view the comparison, then two groups are shown (same-state and cross-state).
- Then the orders in the two groups add up to 96,470.
- Then each group shows its late rate and average delivery days.

US-05 Seller performance (FR-08)

As the Seller Management Lead, I want a ranked list of sellers by late rate, so that I can focus on the sellers with the biggest problem.

- When I open the seller view, then only sellers with at least 30 delivered orders appear.
- Then each seller shows its orders, late rate and average review score.
- When I sort by late rate, then the highest late rate is at the top.

US-06 Delay and customer ratings (FR-09)

As the Customer Experience Manager, I want to see review scores by delay, so that I understand how much lateness hurts satisfaction.

- Given no filters are applied, when I view the delay buckets, then they show on time 88,644 orders (4.29), 1-3 days late 2,662 (3.76), 3-7 days late 1,819 (2.32), 7-14 days late 1,790 (1.75) and over 14 days late 1,555 (1.71).
- Then the bucket orders add up to 96,470.
- Then late orders show an average of 2.57 and on-time orders 4.29.

US-07 Category performance (FR-10)

As the Customer Experience Manager, I want delivery performance by product category, so that I can see whether certain products are delayed more often.

- When I view categories, then names appear in English.
- Then products with no category appear as Unknown.
- Then item revenue across categories adds up to the total item revenue shown on the overview.

US-08 Commercial context (FR-11)

As the Commercial and Finance Analyst, I want order volume and revenue alongside delivery performance, so that I can see what is at stake.

- When I view monthly commercial figures, then revenue is the sum of item price and freight is shown separately.
- Then canceled and unavailable orders are excluded from revenue.
- Then revenue matches the Excel and SQL analysis exactly.

US-09 Filtering (FR-12)

As any dashboard user, I want to filter by month range, state and category, so that I can focus on what matters to me.

- When I change any filter, then every view updates to the filtered data.
- When I clear all filters, then the overall baseline figures return exactly.

US-10 Definitions and limitations (FR-13)

As a member of the Data and BI Team, I want definitions and limitations in the report, so that nobody misreads the figures.

- When I open the definitions page, then each metric definition in section 2 is listed.
- Then the known limitations are listed, including that the promised delivery date looks generous, that months before January 2017 and from September 2018 are excluded, and that review figures cover reviewed orders only.
- Then the number of orders excluded from the delivery base is shown.

6. Traceability

Business requirement	Functional requirements	User stories
BR-01	FR-01, FR-03	US-01
BR-02	FR-02	US-01
BR-03	FR-04, FR-05	US-02
BR-04	FR-06, FR-07	US-03, US-04
BR-05	FR-08	US-05
BR-06	FR-09	US-06
BR-07	FR-10	US-07
BR-08	FR-11	US-08
BR-09	FR-12	US-09
BR-10	FR-03, FR-13	US-01, US-10

Every business requirement has at least one functional requirement and one user story. Section 8 shows which of them were built.

7. Assumptions and Open Items

- The priorities, the speed and usability limits, and the 30-order minimum for sellers are my assumptions. In a real project they would be agreed with stakeholders.
- The same-state rule (D-06) is a simplification for orders with several sellers. In the delivery base, 1,275 orders (1.3%) have items from more than one seller.
- The figures in the acceptance criteria come from the source data. The parts of the dashboard that were built show the same figures.

8. Build Status and Known Gaps

The requirements above describe the full intended solution. Not all of it was built. These requirements are not met in the finished dashboards:

Requirement	Status	What exists instead
FR-05, US-02 (overall late rate as a reference line)	Not built	The monthly trend shows the late rate for each month with no reference line
FR-10, US-07 (late rate by product category)	Not built	Category figures are in the Excel workbook (Pivot_Categories sheet) and the SQL analysis, but not on the dashboards
FR-12, US-09 (filter by product category)	Partly built	Filters for month range and customer state work. A route type filter was added. There is no category filter
FR-02, US-01 (median and 90th percentile delivery days)	Partly built	The Power BI dashboard shows the average only. The median (10.22 days) and 90th percentile (23.1 days) are in the BRD baseline

Requirement	Status	What exists instead
FR-08, US-05 (seller list)	Not on the dashboard	Seller late rates are in the Excel workbook (Pivot_Sellers sheet) and the SQL analysis (sellers with 30 or more orders), but not on the dashboards
FR-13, US-10 (definitions and limitations page)	Partly built	The Power BI dashboard has a short definitions note. The full definitions, data issues and assumptions are in the Excel workbook (Data_Dictionary, Data_Quality_Log and Assumptions_Log sheets), not on a separate dashboard page
FR-11, US-08 (orders and item revenue by month, freight separate)	Not built	Total item revenue and freight are calculated and checked in SQL (02_checks.sql, Check D), but there is no monthly view

The remaining requirements are covered by the Excel workbook, the SQL queries or the Power BI dashboard. Product category is the main gap, and it would be the first addition in a second version.

How this document was prepared

This Requirements Specification was prepared with the assistance of Claude, an AI assistant from Anthropic. Claude made the documentation process smoother by suggesting the industry-standard structure, helping with the wording, and checking that the numbers match across the Excel, SQL and Power BI work. It also answered my questions and pointed out my mistakes. The data analysis in Excel, SQL and Power BI and the dashboard were done by me. Claude helped me with some of the SQL queries, which I ran and checked against my Excel results. The roles, priorities and limits are assumed, because this is a practice case study.